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THE INFLUENCE OF DOSTOEVSKY ON THE CREATIVE WORK OF A. A. UKHTOMSKII

The scientific career of the outstanding Russian physiologist A. A. Ukhtomskii was complex and full of contradictions. A descendant of Prince Vsevolod Big Nest [Bol'shoe Gnezdo] of Suzdal', he was strongly influenced by the traditions and legends of his caste. As a juvenile he was sent to the Nizhnii Novgorod Corps of Cadets where he developed a profound interest in philosophy, psychology, history, and literature. His fellow cadets of the same age were amazed at, and sometimes ridiculed the young Ukhtomskii's attempts to find answers to the burning questions of the day in the books of Shakespeare and Dostoevsky and in scientific works. Ukhtomskii's notebooks and synopses of those years show how this youth, from age 15 to 17, pondered over cardinal problems of world view, the contradictions of Russian society, and the future of science. The barracks life of the cadets, the rigorous schedule and supervision did not suppress Ukhtomskii's persistent desire to get a closer knowledge of the life of Nizhnii Novgorod, a major mill town and river port. He managed to make the acquaintance of members of a group of Marxist workers and

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maintained contact with one of them, M. Ia. Iakovlev ("Arsenii"), for over 40 years.

Abandoning a career in the military, in 1894 Ukhtomskii entered the Department of Literature of the Moscow Religious Academy where he continued his studies of philosophy, psychology, and history. There he made a thorough study of Plato, Aristotle, Seneca, Marcus Aurelius, Augustine, Spinoza, Descartes, and Kant. In his second year there, he wrote a small survey of the opinions of theologians and philosophers on "free will." Problems of ethics took on a new ring for him as a consequence of his personal acquaintance with Leo Tolstoy in the period when the official Church was severely persecuting Tolstoy.

Work on his doctoral dissertation devoted to philosophers' views of space, time, matter, and the evolution of the world inspired Ukhtomskii to make a deep study of the writings of Kant, Hegel, and Newton and to direct attention to problems in the evolution of life. In the synopses one encounters critical comments with respect to the fierce dispute between E. Dubois-Reymond and Ernst Haeckel on the bounds of knowledge and enigmas of the universe, and excerpts from speeches by Helmholtz, from the textbook by the physiologist Max Verworn, and from the books of Darwin and Karl Gegenbaur. However, problems of ethics and morality were not ignored, and the characters of Dostoevsky's books claimed his intense interest.

Ukhtomskii's successful defense of his doctoral dissertation in the fall of 1898, his rejection of an assured career, and his undertaking an ordinary teacher's job in the Volokamsk County School demonstrate that he continued to search for his own road in life. Intensive studies of history, physiology, and philosophy found reflection in various synopses and letters to parents and acquaintances. An important conclusion was arrived at: "We have become accustomed to thinking that physiology is one of the special sciences, necessary to the physician but unnecessary in the development of a world view. But this is untrue. Today it must be understood that separation of 'soul' from 'body' is something that has grounds only in history, that

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the work of the 'soul' — the development of a world view — cannot occur without the knowledge of the 'body,' and that physiology must be classed with the guiding rules in studying the laws of life (in the broad sense)." (1)

In the spring of 1899, Ukhtomskii decided to enter the University of St. Petersburg and become a physiologist. Inspired by the major change about to take place in his life and the fact that he had determined his calling, he wrote the following lines in a notebook, May 25th, 1899: "One cannot but love science, cannot but love the principles by which science lives. One cannot but love Hegel." (2) These few lines alone, written in the months when he had chosen the career of physiologist, testify to the fact that he approached his future work in one of the specialized sciences from a broad philosophical standpoint. He sought in physiology the key to solution of the basic problems of human existence, the key to an understanding of the mechanisms of behavior of the individual and his interrelationships with others. In this connection, Dostoevsky exercised a major influence on him. Dostoevsky's amazingly subtle psychological understanding won Ukhtomskii's mind and emotions early in life, and he was to return many times to the study of the writer's novels and reflection on the lives of his characters. Considerably later, in the spring of 1918, in a letter to V. A. Platonova Ukhtomskii commented: "It is pertinent to ask whether you have read Dostoevsky's 'The Double' [Dvoinik] ? It is a strange story written in his youthful period. I subsequently read with amazement in fragments of Dostoevsky's recently discovered diaries that, in his opinion, the double was one of the most important themes for him, one he agonized over from the earliest years of his writing and to which he returned more than once — in A Raw Youth [Podrostok] and in The Karamazovs [Brat'ia Karamazovy] — and which he almost failed to settle accounts with because of its difficulty.... Long ago, since I was 19 or 20, I began to wonder wherein lay the essence of the problem that tormented Dostoevsky. And it seems to me that that gradually became clear. You know, the hardest thing in the world for a person may be to free himself of his

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double, his automatic tendency to see himself, his faults, his shortcomings, his secret deviance in everyone he meets; but even more difficult is to liberate himself from the constant company of his double! And only from the moment when the double is overcome is the road open to free communication with another!

"It is then for the first time that mind, heart, and word are revealed, making it possible truly to hear what it is that makes the person one has encountered tick.... An open heart and open and intelligent hearing detect and understand in the other that which the individual had been unaware of in himself, and which merely dully exhausted and tormented him from some undiscovered part of his insides! Whereupon people make their way by the thousands to this 'understanding person,' thousands of interlocutors come from all directions to enter into communication with this person who had taught himself to see and hear the people who took the place of his discarded double." (3)

Thirteen years later Ukhtomskii discussed the content of the story "The Double" more fully in a letter to his woman student F. G. Ginzburg, and we regard it as necessary to quote some of his reasoning: "I think that you are discovering what only became clear to me after closer reading of the tale of Mr. Goliadkin. In the final analysis this is a philosophical and psychiatric treatise about solipsism and self-assertion as the basic characteristics of a typical representative of European civilization.... In my opinion, the basic position the author wishes to take and justify here lies in the fact that a basic loneliness, a rationalist egocentrism calls forth, as a direct consequence, constant persecution by one's own image: no matter where an individual looks, no matter whom he encounters, he is fated to see himself because he had accustomed himself to seeing only through himself.... Dostoevsky, it seems, deliberately depicted a petty, insignificant official as bearer of the immense problem of self-assertion. The author wished to emphasize that the problem is not of any 'tremendous' natural gifts in an individual that bring him to 'Napoleonic behavior' or Lermontov's 'demonism' or Nietzsche's 'magnificent beast.'

Dostoevsky wants to emphasize that the most insignificant, ungifted, little European-type person bears the embryo of 'delusions of grandeur' because he has been seized by the 'epidemic of self-affirmation' with a fateful incapacity to see in his neighbor an existence in the world equal in value to his own."

Having analyzed certain passages in "The Double," Ukhtomskii wrote: "It was precisely from this that there was revealed to me the law of the deserved interlocutor as one of the most constant and unavoidable companions of a man along all the paths he travels. To the solipsist the deserved interlocutor is himself, from whom there is nowhere to hide. To a simple and open person, the deserved interlocutor is every individual he meets and every being encountered on the way, who prove in their essence to be precisely what the individual deserved: to a good person, good; to an evil person, evil; to a loving person, loving. It is precisely in this regard that an individual proves to be a powerful civilizing force both to others and to himself." (4)

* * *

In literature we encounter a description of this kind of responsive and sensitive interlocutor and advisor in The Brothers Karamazov in the person of old Father Zosima. In Ukhtomskii's lecture notebook for 1927-1929 there is a significant entry setting forth his intense interest in Zosima in connection with his thinking about the problem of the interlocutor-companion:

"My initial, first, and last task is to understand how the kind of sensitivity possessed by the Elder Zosima comes into being. I learned that it is created by a great physical effort, a tradition handed down from others, and an attitude toward the world as toward a beloved, respected companion with whom one is close to the point of intimacy. This very difficult, constantly tense mode of perception is cultivated and held onto with great effort, with constant self-discipline, and with cautious protection of conscience. But it is of exceptional social value. People cling to an individual who has this quality, apparently

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because an individual cultivated in this way turns out to be uncommonly sensitive and responsive to the lives of other persons and is able readily to place himself in the position of the other, with his attitudes and tribulations. Such an individual is usually less wrapped up in himself and places less emphasis on himself and his infallibility than do others. He is accustomed to criticizing himself constantly and deeply. As a consequence, he is humble within and does not criticize people unless and until they ask his help in their troubles! If he does criticize others, it is like a physician seeking to reason out the disease from which his unfortunate patient suffers...."

Turning from his analysis of Zosima to a characterization of the Karamazovs, Ukhtomskii wrote: "Fiodor Pavlovich, Mitia, and Alexei are all separateness and isolation; no matter which person you examine, he has his own special world, his claims — and this is the origin of his particular unhappiness, his special sin, which disturbs the capacity to live with people! Moreover, the behavior of each is like his consciousness, and his consciousness is like his general attitude. Here each has a closed circle from which it is exceedingly difficult to break out — usually impossible without external assistance! Only if another person is moved and extends patient help, can an individual be torn out of this fateful subject-object relationship, owing to the fact that the world is to each individual what he deserves, and the individual is like his particular world! For one needs nothing more nor less than to change in a person his physiological perception, that which is physiologically customary, the continuity of his life...." "Whatever an individual's dominants [dominanty], such too is his integral image of the world; and whatever his integral image of the world — whatever his behavior — such is his happiness and misfortune — such is the face others see. It would seem that one can say that the elder Zosima and those like him possess a technique for penetrating into the immediate future in a highly unusual way and with regard to the single individual closest to them. They possess a dominant for the personalities of others....

"For Zosima this method is the initial one and fundamental from the outset... and it is a matter of constant tension and effort of an entire life, day after day. The average, calm 'intellectual,' valuing the comfort of self-satisfaction above all else, will hardly take this road! He will always attempt to isolate himself for the sake of his peace in some reassuring and rewarding theory." (5)

These excerpts from Ukhtomskii's notebooks and letters testify to the tremendous influence exercised by Dostoevsky's works on the shaping of his scientific interests and his original interpretation of problems of relations among human beings.

We regard it as necessary to cite yet another excerpt from an addendum he later made at the end of the manuscript of his doctoral dissertation (1898). Its content shows how great a stimulus to his theorizing quests was provided by the creative images, tragic situations, and philosophical aspects of Dostoevsky's works: "It is necessary to develop and continue the empiricism and physiological approach in the study of the routes and norms of man's acquisition of knowledge and good will in the light of the modern data of physiology, psychoanalysis, and epistemology. In particular, Russian literature has given us a precious pioneer along this road in the person of F. M. Dostoevsky."

After analyzing the article on Dostoevsky by V. V. in No. 1 of The Russian Contemporary [Russkii Sovremennik] for 1924 (p. 330), Ukhtomskii wrote: "As soon as one senses the fundamental superiority of living reality and persons over self-asserting, self-isolating knowledge seeking security for itself, one discovers that it is only via culture in the whole man, the culture in his deeds (will and love), that a vision can be achieved in the great gloom of conflicting human souls and persons.

"And it is in this half-light and turmoil of the conflicting ideas and persons that appear in Dostoevsky's works that the contours of three regularities appear: the law of the dominant, the law of the deserved interlocutor-companion, and the law of charity." (6)

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By "the interlocutor" Ukhtomskii understood various types of intercourse between one person and others: contact with an author in reading a work of literature or science, with an artist in viewing his picture and striving to understand the idea it expressed, with a composer in performing his work, with a professor in listening to his lecture, and in conversations with relatives, friends, and acquaintances when an individual faces the task of understanding the inclinations, hopes, thoughts and dreams, emotions of joy and sorrow of other human beings, of obtaining a clear picture of their inner worlds and their struggles, and of discovering something in common and familiar on the plane of intellectual and emotional perception of the world and the events and contradictions within it. "The law of the deserved interlocutor-companion" assures one, Ukhtomskii was convinced, of the opportunity to develop in oneself a "dominant relative to the visage of another person" and thanks to this, to enrich oneself with impressions of various aspects of the internal life of the people of one's milieu, obtaining from this friendly intercourse new impulses for one's own growth, the shaping of ethical principles, and the capacity to cognize little-understood characteristics in oneself through communion with others.

It is no accident that in his textbook, A Sketch of the Physiology of the Nervous System [Ocherk fiziologii nervnoi sistemy], written in besieged Leningrad, Ukhtomskii wrote the following on the significance of the organ of hearing: "Perhaps we can accept hearing as the most important of the sense organs in man, for it is precisely this one particularly that helps man become what he is. The essence of the matter does not lie in music and harmonies or psychological 'emotions' and acoustic impressions .... Human hearing bears an exceptional and most critical responsibility going far beyond the bounds of physiology: that of serving as bulwark and mediator in the great work of organizing speech and verbal intercourse." (7)

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In turning to Dostoevsky for solution of problems of world view and ethics, the young Ukhtomskii at the same time made an intensive effort to understand the meaning of the social processes that had seized hold of Russian society at the turn of the century. He participated in gatherings of revolutionary students in 1901 and 1905, helped a crewman of the "Potemkin," who came from his home district, to hide and flee abroad, and made the acquaintance of Bolshevik students. But for the most part, Ukhtomskii maintained the position of an observer of these tempestuous political events. Recognizing his inability to become a political fighter, he noted on September 28, 1906: "I am a transitional form. I love the old but am no longer capable of living by it. I respect and welcome the new, but have thus far been unable to give myself to it. Life is being made right alongside me, but I cannot overtake it and enter into its splendid movement. It will not wait for such as me." (8)

Tormented by the contradictions of his nature, Ukhtomskii consecrated his talent and capacity for synthesis to science, seeing in it the justification and meaning of his life.

In Ukhtomskii's dissertation, taking Sechenov's reflex theory as point of departure, his earlier critical attitude toward bold transfer of regularities observed in amphibians to the behavior of higher animals and man was overcome.

The period of 1911-1914 was characterized by intensive teaching and theoretical work on Ukhtomskii's part, and if one were to trace, via his synopses and notebooks, the direction of his interests, one might find somewhat surprising his attraction to the new ideas of space and time expressed by Einstein and Minkowski. However, this was not accidental, for as early as 1897 Ukhtomskii, in his work on his doctoral dissertation, tried to interpret the notions of time and space held by philosophers, physicists, and psychologists, and it was not possible for the original thinking of Einstein and Minkowski to escape his attention. Apparently he was one of the few if not the only physiologist in our country who had the penetration to see that relativity theory would not only have to produce a revolution in the realm of physics, but would substantially change the

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concepts of neurophysiologists and psychologists on the reception of time and space. Sherrington's ideas on anticipatory signals of remote receptors (smell, vision, and hearing) and verification of their information by contact receptors, Sechenov's notions about the role of muscular feeling in cognizing time intervals, N. E. Vvedenskii's discoveries and conclusions about the interrelationships between the rhythms of tissues themselves and those of stimuli from the world without were synthesized by Ukhtomskii with his concepts about the "dominant chain of stimuli."

In the course of his theoretical research, Ukhtomskii observed a convergence of the problems of neurophysiology with physicists' and geometricians' teachings about time and space. As early as 1911-1913 one finds in his synopses certain excerpts from the works of Minkowski and Einstein, and in 1921 in his notebook "The Reflex Nature of Cognition" there are thoughts about the significance of relativity theory to the progress of reflex theory. These questions were discussed by him in fuller detail in the form of eighteen theses of an unpublished paper on the chronotope (1924). (9)

After the triumph of the Bolshevik Revolution, Ukhtomskii joined the ranks of the progressive professors of the University of Petrograd. In 1920 he, a former prince, was elected a deputy to the Petrograd Soviet.

One of the greatest achievements of physiology in our country during the Soviet period was Ukhtomskii's study, The Dominant as Working Principle of the Centers of Nervous Activity [Dominanta kak rabochii printsip nervnykh tsentrov] (1923). He ascribed to the principle of the dominant not purely a physiological but a psychophysiological meaning. He regarded the dominant as the physiological basis for motivation of behavior and thought. In creatively developing the reflex concept of behavior, Ukhtomskii pursued his experimental and theoretical research in the direction of convergence with problems of epistemology. Here Lenin's theory of reflection became his methodological point of reference.

It is pertinent to recall that in his article "On Conditioned

Reflex Action" [Ob uslovnootrazhennom deistvii] (1938), Ukhtomskii, treating the development of reflex theory from Descartes to Pavlov, wrote: "Lenin's remarkable theory of reflection poses one problem after another to the physiological theory of reflexes. New reorganizations of the very notion of the 'reflex' are essential in order to expand its application for analytical purposes. There can be no doubt that reflex theory still has a very large and fruitful future, on the road to which Pavlov's work left an exceptional and indelible mark." (10)

Thus we see how complex was the course of the distinguished Soviet physiologist. The questions of a social and ethical nature to which he sought solutions in Dostoevsky led him to work out the theory of the physiological foundation of mental processes; but he found his philosophical bulwark, enabling him to understand the manner in which these processes are determined and their epistemological meaning, in Lenin's theory of reflection.

Notes

1) See V. L. Merkulov, A. A. Ukhtomskii. Ocherk zhizni i nauchnoi deiatel'nosti (1875-1942), USSR Academy of Sciences Press, 1960, p. 27.

2) Ibid., pp. 28, 38, 40.

3) Archives of the USSR Academy of Sciences, Fond 749, List 1, Folder 95, Sheet of draft of letter to V. A. Platonova.

4) Cited from a copy of a letter from Ukhtomskii to Ginzburg, who retains the original.

5) Archives of the USSR Academy of Sciences, Fond 749, List 1, Folder 91, Lecture Notebook No. 18, pp. 52-54, undated but approximately from 1927-1929.

6) Ibid. Folder 147, Notebook No. 5, p. 104, addendum in green ink, undated but not earlier than 1924.

7) A. A. Ukhtomskii, Sobr. soch., Vol. IV, Leningrad University Press, 1945, pp. 219-220.

8) See V. L. Merkulov, op. cit., p. 37.

9) See V. L. Merkulov, "Printsip dominanty i predstavleniia A. A. Ukhtomskogo o khronotope (vremenno-prostranstvennom

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komplekse)," Uspekhi sovremennoi biologii, 1959, Vol. XLVII, No. 2, pp. 204-205 and 210-218.

10) A. A. Ukhtomskii, a collection entitled Dominanta, Leningrad, "Nauka" Publishing House, 1966, pp. 174 and 177.

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